

FiberDrops Gradient: A Tubular System for Visual Color Gradients via Droplet Density Control

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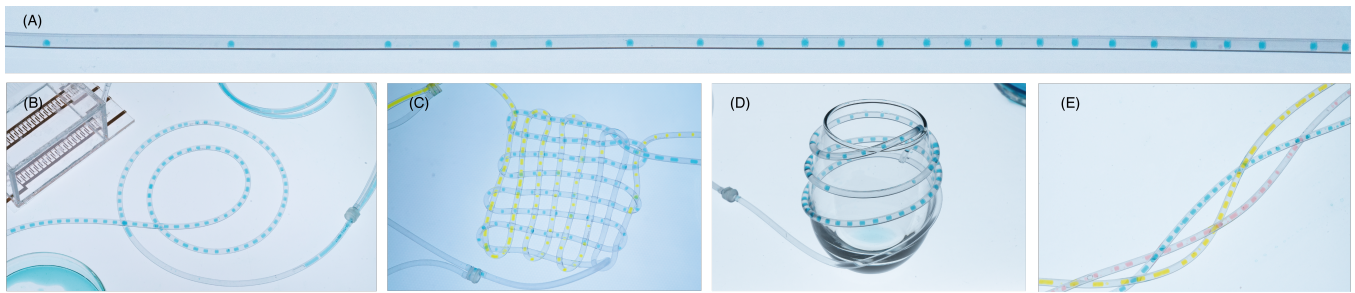


Figure 1: (A) Changing droplet density in a tube. (B) Overall system. (C) Woven tubes. (D) A tube wound around a 3D object. (E) Multi-color expression.

Abstract

Flowing colored liquids in tubes have attracted interest in HCI due to their deformability and programmability, enabling dynamic visual expression. However, intricate patterns such as color gradients remain underexplored. We propose a method to express gradients by controlling droplet density using a custom connector inspired by microfluidic techniques. Fluids are driven by electrohydrodynamic (EHD) pumps, offering a lightweight, compact, and silent alternative to mechanical systems. This approach expands the expressive potential of tubular systems.

Keywords

Programmable Materials, Fluid Interface, Liquid Actuation, Droplet generation, Electrohydrodynamics

ACM Reference Format:

Tomoka Kurosawa, Takafumi Morita, Yumi Nishihara, Shingo Maeda, and Yasuaki Kakehi. 2025. FiberDrops Gradient: A Tubular System for Visual Color Gradients via Droplet Density Control. In *SIGGRAPH Asia 2025 Emerging Technologies (SA Emerging Technologies '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3757373.3763773>

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SA Emerging Technologies '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2133-5/25/12

<https://doi.org/10.1145/3757373.3763773>

1 Introduction

In recent years, liquids have attracted growing attention in both the art and human-computer interaction (HCI) domains due to their high responsiveness to external stimuli [Lu et al. 2021; Matsunobu and Kakehi 2018; Omoya and Hara Design Institute 2009; Yamamoto et al. 2020]. In particular, flowing colored liquids in tubes have been explored as an interface for visual expression. This is enabled by the deformability of the tubular structures and the programmability of the internal fluids, which together support the dynamic display of visual information.

Using mechanical components such as solenoid valves and pumps to insert liquids into tubes and control their flow, previous research has explored both display systems and artistic expressions. [Liu et al. 2022; Manganiello and Goldman 2020; Popp 2011; Sugitani 2020; Suzunaga et al. 2020]. However, intricate visual expressions such as droplet-based color gradients have yet to be fully explored. In the field of painting, the expression of gradational colors has long been explored not only through the selection of hues, but also by varying the density of brushstrokes such as pointillism [Düchting 2000]. Inspired by this painting approach, we propose a method for expressing color gradients in tubes by generating droplets and controlling their density. To drive the liquid, we utilize EHD pumps based on the EHD phenomenon [Pickard 1963a,b; Richter and Sandmaier 1990]. It can generate fluid flow by applying high voltage to the electrodes in a dielectric liquid. Unlike conventional liquid systems, EHD pumps can realize a compact, lightweight, and silent liquid system. This technology has recently gained attention in the field of soft robotics [Cacucciolo et al. 2019; Kuwajima et al.

2022] and HCI [Fujii et al. 2021; Jiang et al. 2023; Kurosawa et al. 2025; Morita et al. 2023; Morita and Kakehi 2023; Morita et al. 2025, 2022]. Liquebits [Aoyama et al. 2025] is a notable example exploring colored liquid expressions in tubes using EHD pumps and focuses mainly on the silent expression and does not address color gradients. Although they alternately control the pumps at the determined time intervals to generate droplets, in this research, we modulate the droplet density by adjusting the output of the EHD pumps.

2 FiberDrops Gradient

In this study, the transparent liquid is defined as the continuous phase, while the colored liquid that is formed into droplets is defined as the dispersed phase. The flows are driven by EHD pumps, and droplets are generated when the continuous phase shears and breaks the flow of the dispersed phase. The output of EHD pumps depends on the strength of the electric field between the electrodes. With the electric field strength of the continuous-phase pump held constant, increasing the electric field strength of the dispersed-phase pump leads to a correspondingly higher droplet density.

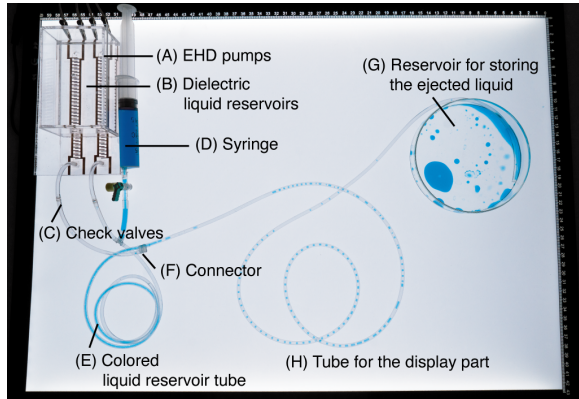


Figure 2: System overview.

As shown in Fig. 2, the system is broadly composed of two components: the liquid actuation part and the display part. The liquid actuation part consists of EHD pumps, the high voltage control unit, dielectric liquid reservoirs, check valves, syringe, colored liquid reservoir tube, connector, and reservoir to store ejected liquid. The liquid flow is generated by the EHD pumps which is controlled by the high voltage control unit. Novec 7100 was used as the dielectric transparent liquid. When the pump output was set to 50%, the flow rate was approximately 63.4 ml/min. Based on this reference point, the voltages corresponding to 50%–90% of the pump output were applied. As the droplets are created by using two immiscible liquids, we used Novec 7100 and colored water. The display part which consists of a tube can be deformed due to the flexibility of the material. For the tubes we used Tygon® tubing with an inner diameter of 2 mm and an outer diameter of 4 mm. The outer diameter of the connector's inlets and outlets is determined by the inner diameter of the tube. Since all connectors are 3D printed, their size can be customized for the intended applications. However, the narrower the diameter, the higher the internal resistance in the tube, requiring appropriate adjustment of the pump output. Before activating

this system, the colored water containing syringe is pushed to load it into the reservoir tube. Then, high voltage is applied to both EHD pumps; one pump drives the dielectric liquid as the continuous phase, while the other controls the dielectric liquid to push the colored water towards and through the connector. The two immiscible liquids meet within the connector, and droplets are formed as they are pushed into the display tube. To prevent backflow, check valves are installed in the pump system. By adjusting the balance of the pumps' output, the density of the droplets can be modulated, enabling the expression of color gradients using liquid.

As shown in Fig. 3, the connector applies the co-flow configuration, a common droplet generation method in microfluidics [Cramer et al. 2004]. The connector consists of two inlets, a chamber and an outlet, where each liquid enters its assigned inlet, merges within the chamber, and exits from the outlet. The colored liquid is segmented into droplets by the combined effects of shear stress exerted by the continuous phase and the interfacial tension between the two immiscible liquids. The connector was fabricated with an SLA printer (Form 2, Formlabs Inc.) using clear resin and assembled using UV-curable adhesive.

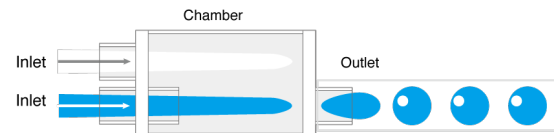


Figure 3: Inner structure of the connector.

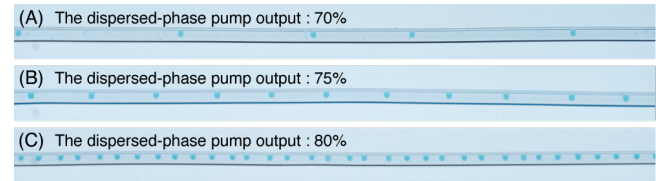


Figure 4: Result.

Figure 4 shows that the droplet density increases as the electric field strength of the dispersed-phase pump increases, with the continuous-phase pump output set to 50%. Building on this capability, Fig. 1(C)–(E) illustrate design examples that demonstrate the potential applications of the system.

3 Conclusion

In this paper, we proposed a novel method for controlling droplet density to express gradients. Looking ahead, we aim to display gradient-based patterns that interactively change in response to user's behaviors, and to seamlessly integrate this system into soft materials such as clothing and bags to enable dynamic pattern expression.

Acknowledgments

We sincerely thank Professor Shunji Yamanaka for his insightful advice and warm support. This work was supported by JSPS KAKENHI Grant Number JP24H00708, JST SPRING Grant Number JPMJSP2108 and JST ACT-X Grant Number JPMJAX22KI.

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